11/10/25

# EXPERIMENT-8

AIM: To discover live hosts using Nmap Scans (ARP, ICMP, TCP/UDP) on the TryHackMe platform Room Link: https://tryhackme.com/room/nmapst

Intro:

Nmap is a free open source network scanning tool used to identify active hosts and services It was developed by Gordon Lyon And released in 1997

Nmap Live Host Discovery identifies online systems before port Scanning reducing unnecessary scanning. It uses ARP Scans, ICMP Scans and TCP/UDP ping scans. to detect live hosts. Nmap also supports network mapping, service detection and scripting for advanced security analysis.

Subnetwork:

A network segment is a group of computers connected through a switch or Wifi access point. A Subnetwork is a logical network with its own IP address range connected to other networks through a router

/16 subnet subnet mask 255.255.0.0

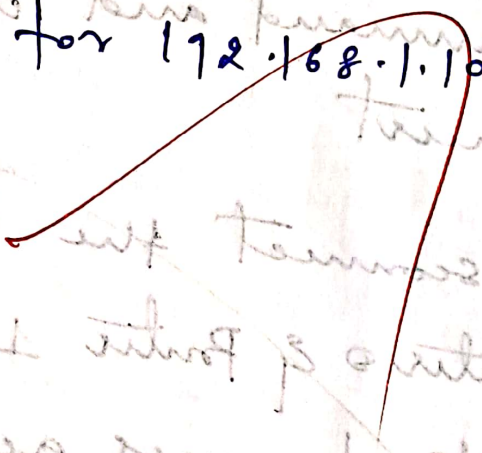
/24 subnet subnet mask 255.255.0.0

During Nmap uses ARP to discover live hosts on the same subnet by finding their MAC addresses ARP request cannot cross routers. So.

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Output

Nmap Scan report for 192.168.1.10



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## Enumerating Targets

Nmap accepts targets as individual IP addresses ranges, or subnets. The -iL option reads targets from a file -sL lists target without scanning and -n disables DNS resolution.

IP for 10.10.12.18/29 : 10.10.12.8.

Number IP add: 10.10.0 – 255.101-125:6400

## TCP/IP Layers

ARP : Link
ICMP : Network
TCP : Transport
UDP : Transport

Before pinging a host on the same subnet a computer sends an ARP Request and receives an ARP Response to obtain the destinations MAC address.

## NMap Host Discovery Using ARP

Nmap uses ARP requests to discover live hosts on the local ~~hosts~~ subnet. It is effective for local network discovery because ARP does not cross routers

nmap -PR -sn TARGET

## Nmap Host Discovery Using ICMP

Nmap uses ICMP packets to identify active hosts. Firewalls may block ICMP Echo Requests.

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Nmap Host Discovery Using Tcp and UDP

Tcp SYN ping : Sends a SYN packets

Tcp ACK ping : Sends a ACK packets

UDP ping : sends UDP packets.

Reverse DNS Lookup

Reverse DNS Lookup identifies hostnames associated with Ip addresses

-n - disables DNS resolution

-R - forces reverse DNS lookup for targets

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# EXPERIMENT-8B

AIM: b) Configuration of wireless LAN using Cisco Packet Tracer.

PROCEDURE

1. place a linksys wireless router and three wireless PCs in Cisco Packet Tracer.
2. Open the router go to Administration set the username and password to admin and save the settings.
3. Under Wireless set the SSID to MotherNetwork
4. Under Wireless Security select WEP and set key to 0123456789 Save the settings.
5. Configure static IP addresses on the pcs under Desktop → IP Configuration

| Device | IP Address | Subnet Mask | Default Gateway |
|---|---|---|---|
| Router | 192.168.0.1 | 255.255.255.0 | — |
| PC0 | 192.168.0.2 | 255.255.255.0 | 192.168.0.1 |
| PC1 | 192.168.0.3 | 255.255.255.0 | 192.168.0.1 |
| PC2 | 192.168.0.4 | 255.255.255.0 | 192.168.0.1 |

6. On each pc open desktop → pc wireless → Connect and click Refresh.
7. Select MotherNetwork click connect enter the security key 0123456789 and connect
8. Repeat the process for all three pcs
9. Result: A wireless LAN is configured successfully.

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